

Naïve Bayes Classifier

Bayes' Rule

$$P(\text{hypothesis}|\text{evidence}) = \frac{P(\text{evidence}|\text{hypothesis}) \times P(\text{hypothesis})}{P(\text{evidence})}$$

Equivalently

$$P(Y = y_k | X = \mathbf{x}_i) = \frac{P(X = \mathbf{x}_i | Y = y_k) P(Y = y_k)}{P(X = \mathbf{x}_i)}$$

where X is a R.V. representing the evidence and Y is a R.V. for the class label (hypothesis).

$$P(Y = y_k | X = \mathbf{x}_i) = \frac{P(Y = y_k) P(X_1 = \mathbf{x}_{i,1} \cap \cdots \cap X_d = \mathbf{x}_{i,d} | Y = y_k)}{P(X_1 = \mathbf{x}_{i,1} \cap \cdots \cap X_d = \mathbf{x}_{i,d})}$$

where X_j denotes the random variable for the j th feature.

Naïve Bayes Classifier

- Idea: Use the training data to estimate $P(X|Y)$ and $P(Y)$.
- Use the Bayes' rule to infer $P(Y|X_{new})$ for new data.

$$P(Y = y_k | X = \mathbf{x}_i) = \frac{P(Y = y_k)P(X_1 = \mathbf{x}_{i,1} \cap \cdots \cap X_d = \mathbf{x}_{i,d} | Y = y_k)}{P(X_1 = \mathbf{x}_{i,1} \cap \cdots \cap X_d = \mathbf{x}_{i,d})}$$

- $P(Y = y_k)$: easy to estimate from the data
- $P(X_1 = \mathbf{x}_{i,1} \cap \cdots \cap X_d = \mathbf{x}_{i,d})$: Determination is not necessary
- $P(X_1 = \mathbf{x}_{i,1} \cap \cdots \cap X_d = \mathbf{x}_{i,d} | Y = y_k) = P(X_1, X_2, \dots, X_d | Y)$: Estimation is impractical but necessary

Naïve Bayes Classifier

- Issue: Estimating the joint PD is not practical
- If we assume that the features are independent given the class labels

$$P(X_1, X_2, \dots, X_d | Y) = \prod_{j=1}^d P(X_j | Y)$$

- We assume all the features are conditionally independent given Y
- This is often violated in practice

Training Naïve Bayes Classifier

- Estimate $P(X_j|Y)$ and $P(Y)$ from the training data by counting process.

x_1 <u>Sky</u>	x_2 <u>Temp</u>	x_3 <u>Humid</u>	x_4 <u>Wind</u>	x_5 <u>Water</u>	x_6 <u>Forecast</u>	Y <u>Play?</u>
sunny	warm	normal	strong	warm	same	yes
sunny	warm	high	strong	warm	same	yes
rainy	cold	high	strong	warm	change	no
sunny	warm	high	strong	cool	change	yes

$$\frac{\#(Sky = \text{sunny} \wedge \text{play})}{\# \text{play}}$$

$$P(\text{play}) = ?$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = ?$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = ?$$

...

$$P(\neg \text{play}) = ?$$

$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = ?$$

$$P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

...

Training Naïve Bayes Classifier

- Estimate $P(X_j|Y)$ and $P(Y)$ from the training data by counting process.

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Wind</u>	<u>Water</u>	<u>Forecast</u>	<u>Play?</u>
sunny	warm	normal	strong	warm	same	yes
sunny	warm	high	strong	warm	same	yes
rainy	cold	high	strong	warm	change	no
sunny	warm	high	strong	cool	change	yes

$P(\text{play}) = ?$

$P(\text{Sky} = \text{sunny} \mid \text{play}) = ?$

$P(\text{Humid} = \text{high} \mid \text{play}) = ?$

...

$P(\neg \text{play}) = ?$

$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = ?$

$P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$

...

Training Naïve Bayes Classifier

- Estimate $P(X_j|Y)$ and $P(Y)$ from the training data by counting process.

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Wind</u>	<u>Water</u>	<u>Forecast</u>	<u>Play?</u>
sunny	warm	normal	strong	warm	same	yes
sunny	warm	high	strong	warm	same	yes
rainy	cold	high	strong	warm	change	no
sunny	warm	high	strong	cool	change	yes

$$P(\text{play}) = 3/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = ?$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = ?$$

...

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = ?$$

$$P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

...

Training Naïve Bayes Classifier

- Estimate $P(X_j|Y)$ and $P(Y)$ from the training data by counting process.

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Wind</u>	<u>Water</u>	<u>Forecast</u>	<u>Play?</u>
sunny						yes
sunny						yes
rainy	cold	high	strong	warm	change	no
sunny						yes

$$P(\text{play}) = 3/4$$

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$$P(\text{Humid} = \text{high} \mid \text{play}) = ?$$

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$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = ?$$

$$P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

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<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Wind</u>	<u>Water</u>	<u>Forecast</u>	<u>Play?</u>
sunny						yes
sunny						yes
rainy	cold	high	strong	warm	change	no
sunny						yes

$$P(\text{play}) = 3/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = 1$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = ?$$

...

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = ?$$

$$P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

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Training Naïve Bayes Classifier

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$$P(\neg \text{play}) = 1/4$$

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Training Naïve Bayes Classifier

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		high				yes
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		high				yes

$$P(\text{play}) = 3/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = 1$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = ?$$

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$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = 0$$

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		normal				yes
		high				yes
rainy	cold	high	strong	warm	change	no
		high				yes

$$P(\text{play}) = 3/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = 1$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = 2/3$$

...

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = 0$$

$$P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

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Training Naïve Bayes Classifier

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		high				no
sunny	warm	high	strong	cool	change	yes

$$P(\text{play}) = 3/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = 1$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = 2/3$$

...

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = 0$$

$$P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

...

Training Naïve Bayes Classifier

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$$P(\text{Humid} = \text{high} \mid \text{play}) = 2/3$$

...

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = 0$$

$$P(\text{Humid} = \text{high} \mid \neg \text{play}) = 1$$

...

Laplace Smoothing

- Some probabilities estimated by counting process might be zero—possible case of overfitting!

- Regularize by Laplace smoothing

- Add 1 to each count

$$P(X_j = v | Y = y_k) = \frac{c_v + 1}{\sum_{v' \in \text{values}(X_j)} c_{v'} + |\text{values}(X_j)|}$$

- c_v is the count of training examples with a value of v for feature j and class label y_k
 - $\sum_{v' \in \text{values}(X_j)} c_{v'}$ = Count of values in that class
 - $|\text{values}(X_j)|$ is the number of values X_j can take on

$$\frac{c_v}{\sum_{v'} c_{v'}}$$

Training Naïve Bayes with Laplace Smoothing

- Estimate $P(X_j|Y)$ and $P(Y)$ from the training data by counting with Laplace smoothing

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Wind</u>	<u>Water</u>	<u>Forecast</u>	<u>Play?</u>
sunny						yes
sunny						yes
rainy	cold	high	strong	warm	change	no
sunny						yes

$$\frac{3+1}{3+2} = \frac{4}{5}$$

$$P(\text{play}) = 3/4$$

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = 4/5$$

$$P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = ?$$

$$\frac{0+1}{1+2} = \frac{1}{3}$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = ?$$

$$P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

...

...

Training Naïve Bayes with Laplace Smoothing

- Estimate $P(X_j|Y)$ and $P(Y)$ from the training data by counting with Laplace smoothing

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Wind</u>	<u>Water</u>	<u>Forecast</u>	<u>Play?</u>
sunny	warm	normal	strong	warm	same	yes
sunny	warm	high	strong	warm	same	yes
rainy						no
sunny	warm	high	strong	cool	change	yes

$$P(\text{play}) = 3/4$$

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = 4/5 \quad P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = 1/3$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = ? \quad P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

...

...

Training Naïve Bayes with Laplace Smoothing

- Estimate $P(X_j|Y)$ and $P(Y)$ from the training data by counting with Laplace smoothing

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Wind</u>	<u>Water</u>	<u>Forecast</u>	<u>Play?</u>
		normal				yes
		high				yes
rainy	cold	high	strong	warm	change	no
		high				yes

$$P(\text{play}) = 3/4$$

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = 4/5 \quad P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = 1/3$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = 3/5 \quad P(\text{Humid} = \text{high} \mid \neg \text{play}) = ?$$

...

...

Training Naïve Bayes with Laplace Smoothing

- Estimate $P(X_j|Y)$ and $P(Y)$ from the training data by counting with Laplace smoothing

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Wind</u>	<u>Water</u>	<u>Forecast</u>	<u>Play?</u>
sunny	warm	normal	strong	warm	same	yes
sunny	warm	high	strong	warm	same	yes
		high				no
sunny	warm	high	strong	cool	change	yes

$$P(\text{play}) = 3/4$$

$$P(\neg \text{play}) = 1/4$$

$$P(\text{Sky} = \text{sunny} \mid \text{play}) = 4/5 \quad P(\text{Sky} = \text{sunny} \mid \neg \text{play}) = 1/3$$

$$P(\text{Humid} = \text{high} \mid \text{play}) = 3/5 \quad P(\text{Humid} = \text{high} \mid \neg \text{play}) = 2/3$$

...

...

Using the Classifier

- We have $P(Y = y_k | X = \mathbf{x}_i) = \frac{P(X=\mathbf{x}_i|Y=y_k)P(Y=y_k)}{P(X=\mathbf{x}_i)}$
- $P(X = \mathbf{x}_i)$ is a constant for a given example (not required for prediction)
- To classify a new point $\mathbf{x}$

$$h(\mathbf{x}) = \arg \max_{y_k} P(Y = y_k) \prod_{j=1}^d P(X_j = \mathbf{x}_j | Y = y_k)$$

$$h(\mathbf{x}) = \arg \max_{y_k} \log P(Y = y_k) + \sum_{j=1}^d \log P(X_j = \mathbf{x}_j | Y = y_k)$$

The Naïve Bayes Classifier Algorithm

- For each class y_k
 - Estimate $P(Y = y_k)$ from the data
 - For each value $x_{i,j}$ of each feature X_i -- Estimate $P(X_i = x_{i,j} | Y = y_k)$
- Classify a new point using:

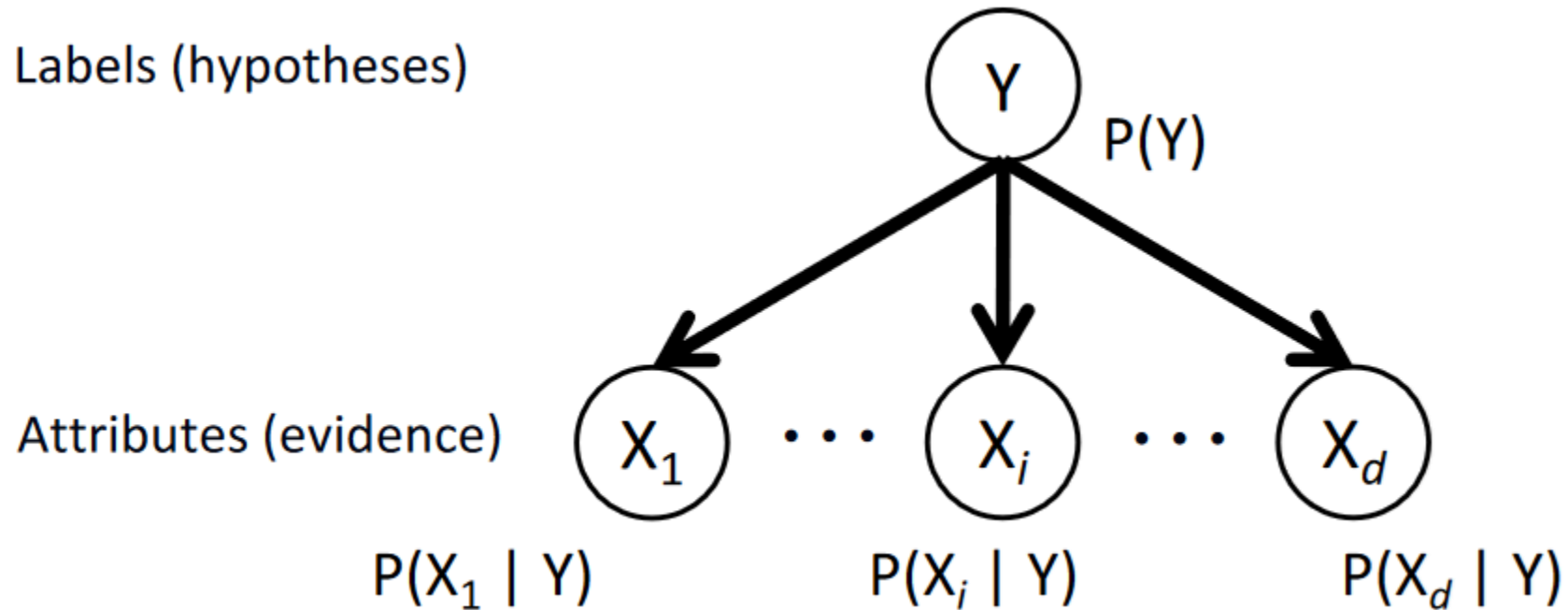
~~Tricky~~

$$h(x) = \arg \max_{y_k} \log P(Y = y_k) + \sum_{j=1}^d \log P(X_j = \mathbf{x}_j | Y = y_k)$$

- Although the independence assumption does not often hold true, Naïve Bayes Classifier performs quite well.

Graphical Model

Labels (hypotheses)

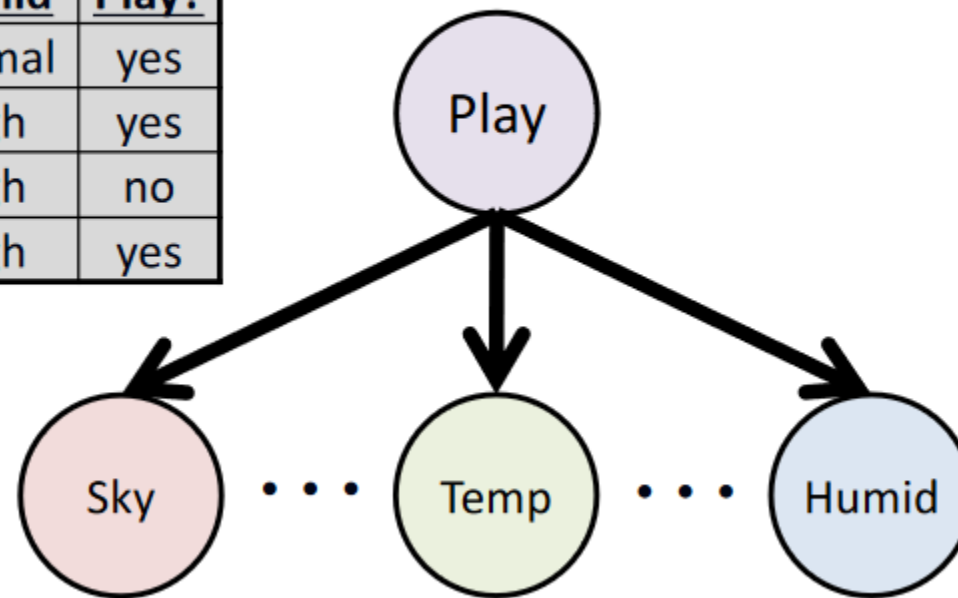


- Nodes denote random variables
- Edges denote dependency
- Each node is associated to its parents via conditional probability

Example

Data:

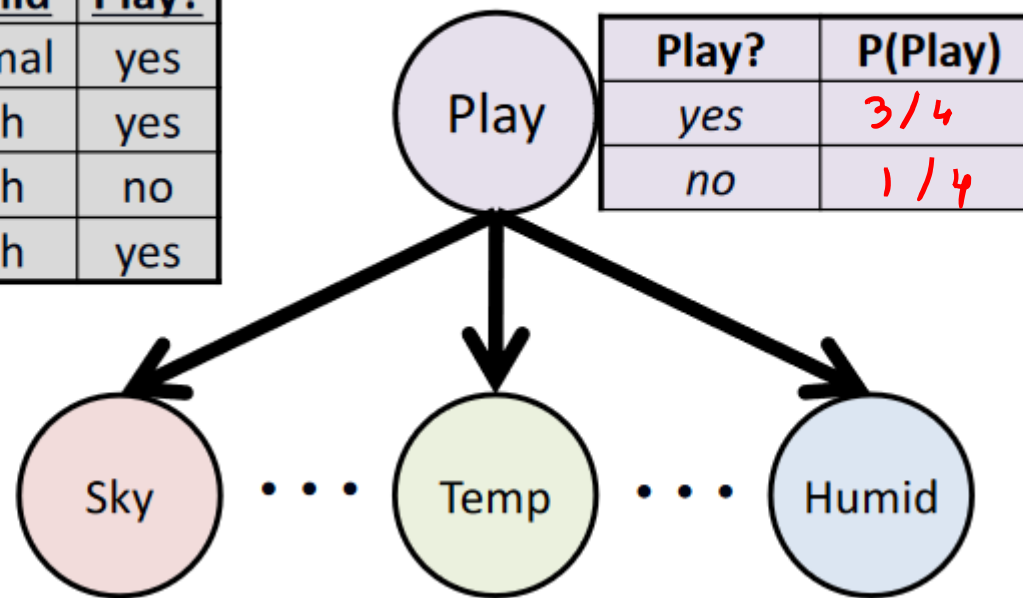
<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Play?</u>
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes



Example

Data:

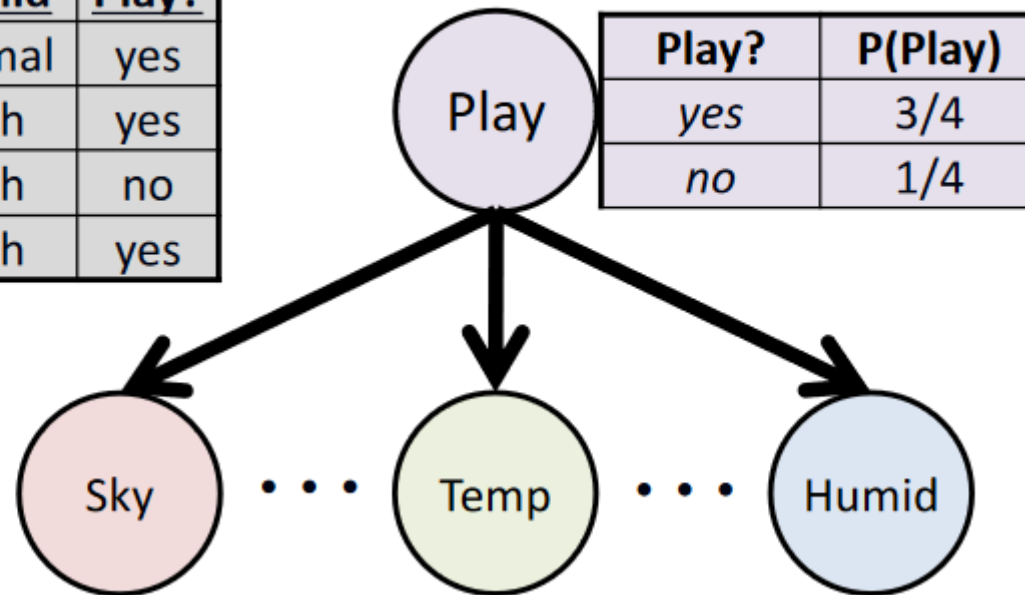
<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Play?</u>
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes



Example

Data:

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Play?</u>
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes

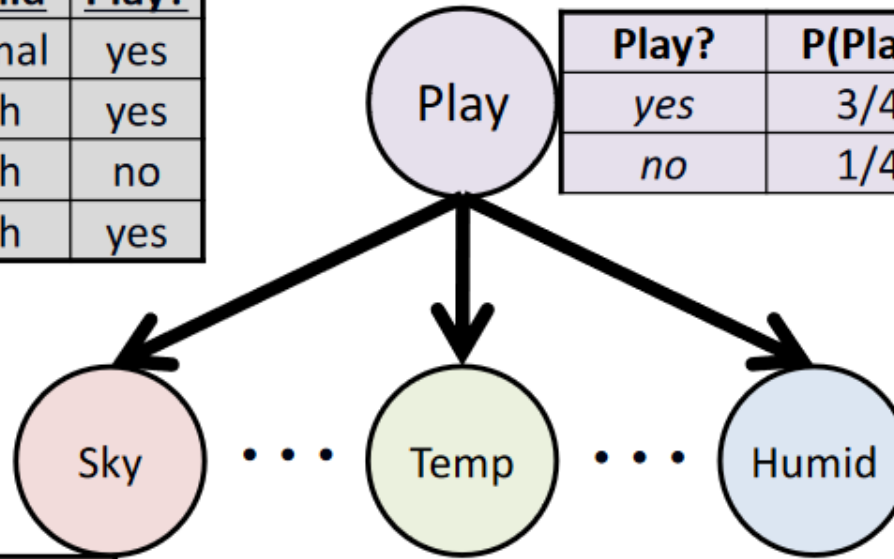


Example

Data:

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Play?</u>
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes

Play?	P(Play)
<i>yes</i>	3/4
<i>no</i>	1/4

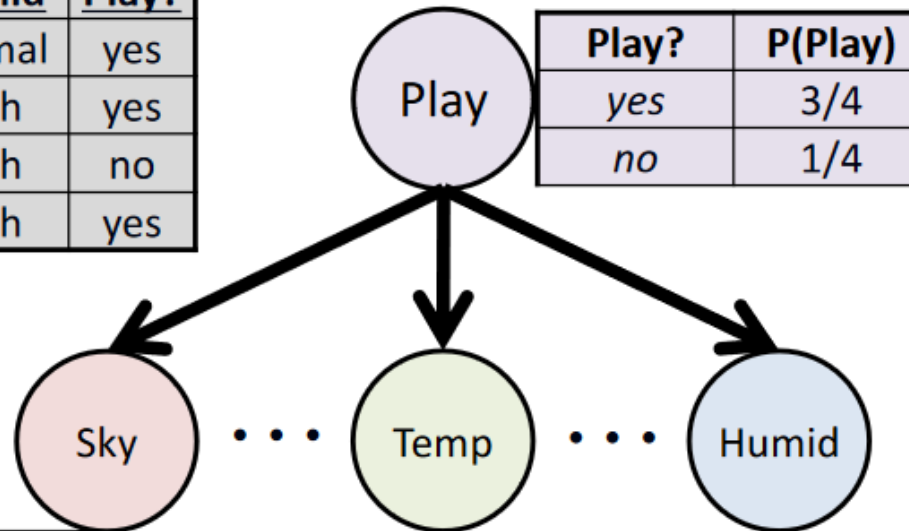


Sky	Play?	P(Sky Play)
<i>sunny</i>	<i>yes</i>	<i>4/5</i>
<i>rainy</i>	<i>yes</i>	<i>1/5</i>
<i>sunny</i>	<i>no</i>	<i>1/3</i>
<i>rainy</i>	<i>no</i>	<i>2/3</i>

Example

Data:

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Play?</u>
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes



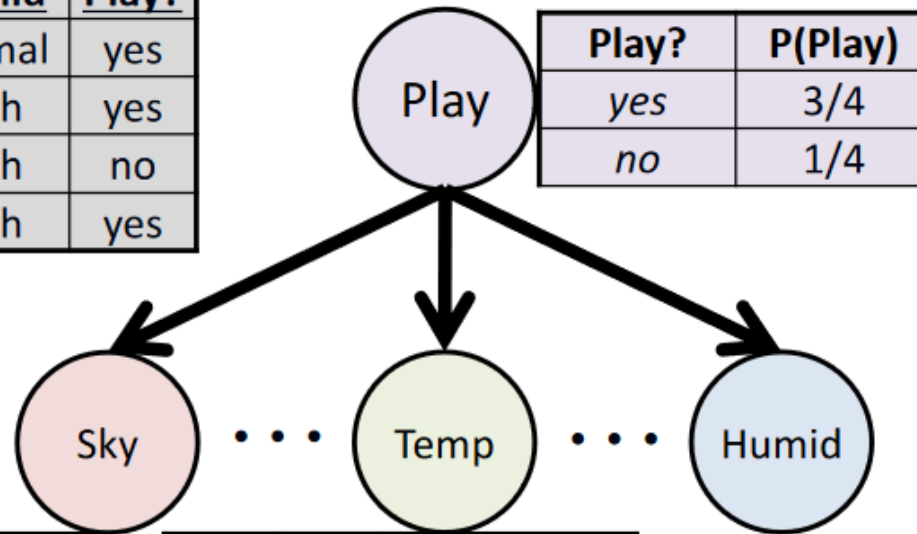
<u>Play?</u>	<u>P(Play)</u>
<i>yes</i>	3/4
<i>no</i>	1/4

<u>Sky</u>	<u>Play?</u>	<u>P(Sky Play)</u>
<i>sunny</i>	<i>yes</i>	4/5
<i>rainy</i>	<i>yes</i>	1/5
<i>sunny</i>	<i>no</i>	1/3
<i>rainy</i>	<i>no</i>	2/3

Example

Data:

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Play?</u>
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes



<u>Play?</u>	<u>P(Play)</u>
yes	3/4
no	1/4

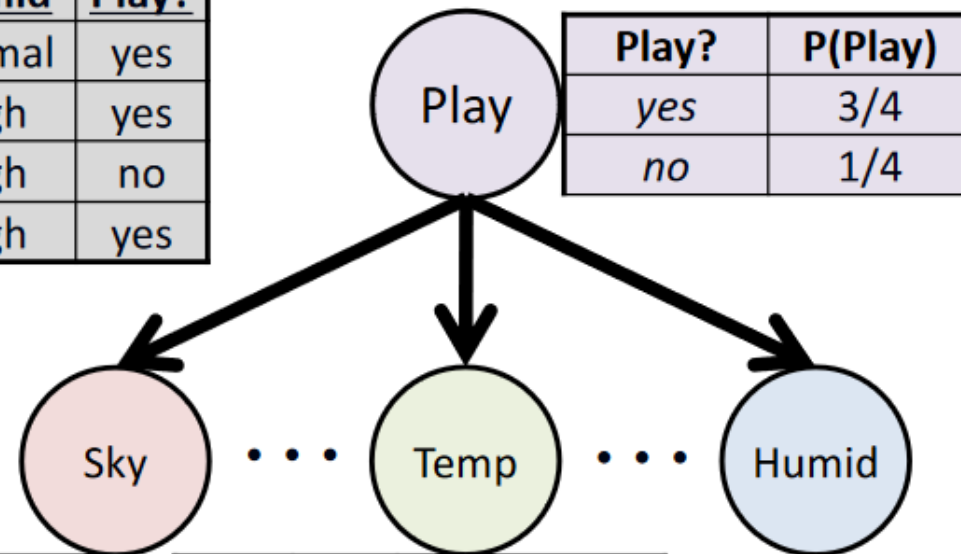
<u>Sky</u>	<u>Play?</u>	<u>P(Sky Play)</u>
sunny	yes	4/5
rainy	yes	1/5
sunny	no	1/3
rainy	no	2/3

<u>Temp</u>	<u>Play?</u>	<u>P(Temp Play)</u>
warm	yes	4/5
cold	yes	1/5
warm	no	1/3
cold	no	2/3

Example

Data:

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Play?</u>
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes



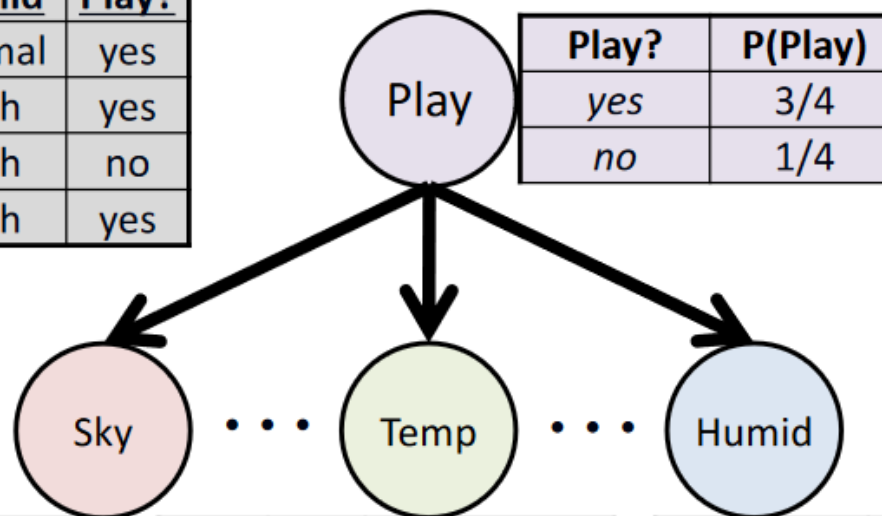
<u>Sky</u>	<u>Play?</u>	<u>P(Sky Play)</u>
sunny	yes	4/5
rainy	yes	1/5
sunny	no	1/3
rainy	no	2/3

<u>Temp</u>	<u>Play?</u>	<u>P(Temp Play)</u>
warm	yes	4/5
cold	yes	1/5
warm	no	1/3
cold	no	2/3

Example

Data:

Sky	Temp	Humid	Play?
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes



Play?	P(Play)
yes	3/4
no	1/4

Sky	Play?	P(Sky Play)
sunny	yes	4/5
rainy	yes	1/5
sunny	no	1/3
rainy	no	2/3

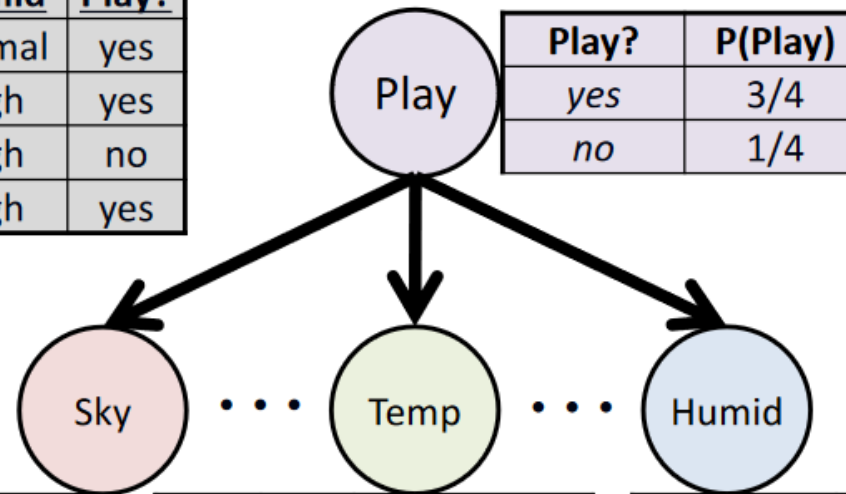
Temp	Play?	P(Temp Play)
warm	yes	4/5
cold	yes	1/5
warm	no	1/3
cold	no	2/3

Humid	Play?	P(Humid Play)
high	yes	3/5
norm	yes	2/5
high	no	2/3
norm	no	1/3

Example

Data:

<u>Sky</u>	<u>Temp</u>	<u>Humid</u>	<u>Play?</u>
sunny	warm	normal	yes
sunny	warm	high	yes
rainy	cold	high	no
sunny	warm	high	yes



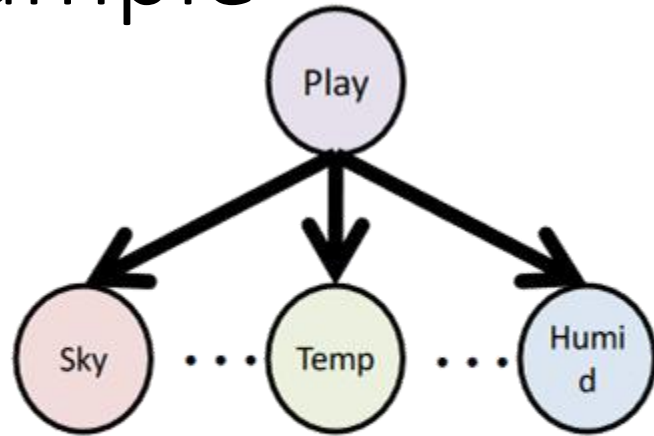
<u>Play?</u>	<u>P(Play)</u>
<i>yes</i>	3/4
<i>no</i>	1/4

<u>Sky</u>	<u>Play?</u>	<u>P(Sky Play)</u>
<i>sunny</i>	<i>yes</i>	4/5
<i>rainy</i>	<i>yes</i>	1/5
<i>sunny</i>	<i>no</i>	1/3
<i>rainy</i>	<i>no</i>	2/3

<u>Temp</u>	<u>Play?</u>	<u>P(Temp Play)</u>
<i>warm</i>	<i>yes</i>	4/5
<i>cold</i>	<i>yes</i>	1/5
<i>warm</i>	<i>no</i>	1/3
<i>cold</i>	<i>no</i>	2/3

<u>Humid</u>	<u>Play?</u>	<u>P(Humid Play)</u>
<i>high</i>	<i>yes</i>	3/5
<i>norm</i>	<i>yes</i>	2/5
<i>high</i>	<i>no</i>	2/3
<i>norm</i>	<i>no</i>	1/3

Example



Play?	P(Play)
<i>yes</i>	3/4
<i>no</i>	1/4

Temp	Play?	P(Temp Play)
<i>warm</i>	<i>yes</i>	4/5
<i>cold</i>	<i>yes</i>	1/5
<i>warm</i>	<i>no</i>	1/3
<i>cold</i>	<i>no</i>	2/3

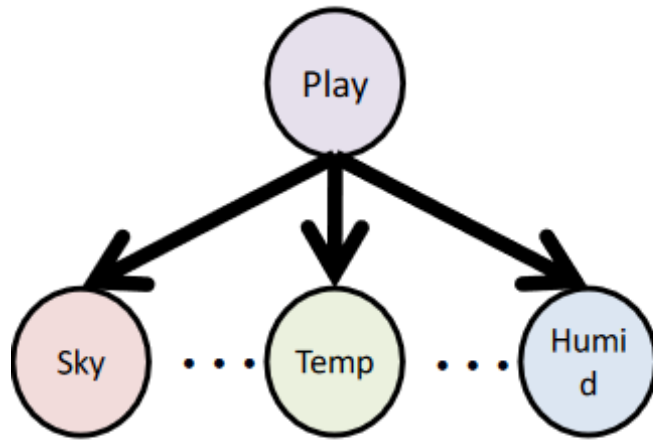
Sky	Play?	P(Sky Play)
<i>sunny</i>	<i>yes</i>	4/5
<i>rainy</i>	<i>yes</i>	1/5
<i>sunny</i>	<i>no</i>	1/3
<i>rainy</i>	<i>no</i>	2/3

Humid	Play?	P(Humid Play)
<i>high</i>	<i>yes</i>	3/5
<i>norm</i>	<i>yes</i>	2/5
<i>high</i>	<i>no</i>	2/3
<i>norm</i>	<i>no</i>	1/3

$$h(x) = \arg \max_{y_k} \log P(Y = y_k) + \sum_{j=1}^d \log P(X_j = x_j | Y = y_k)$$

Objective: Predict label for $x = (rainy, warm, normal)$

Example



Predict label for:
 $\mathbf{x} = (\text{rainy}, \text{warm}, \text{normal})$

Play?	P(Play)
yes	3/4
no	1/4

Temp	Play?	P(Temp Play)
warm	yes	4/5
cold	yes	1/5
warm	no	1/3
cold	no	2/3

Sky	Play?	P(Sky Play)
sunny	yes	4/5
rainy	yes	1/5
sunny	no	1/3
rainy	no	2/3

Humid	Play?	P(Humid Play)
high	yes	3/5
norm	yes	2/5
high	no	2/3
norm	no	1/3

$$\begin{aligned}
 P(\text{play} \mid \mathbf{x}) &\propto \log P(\text{play}) + \log P(\text{rainy} \mid \text{play}) + \log P(\text{warm} \mid \text{play}) + \log P(\text{normal} \mid \text{play}) \\
 &\propto \log 3/4 + \log 1/5 + \log 4/5 + \log 2/5 = -1.319 \quad \text{predict PLAY}
 \end{aligned}$$

$$\begin{aligned}
 P(\neg \text{play} \mid \mathbf{x}) &\propto \log P(\neg \text{play}) + \log P(\text{rainy} \mid \neg \text{play}) + \log P(\text{warm} \mid \neg \text{play}) + \log P(\text{normal} \mid \neg \text{play}) \\
 &\propto \log 1/4 + \log 2/3 + \log 1/3 + \log 1/3 = -1.732
 \end{aligned}$$

Summary

- **Advantages**

- Fast training
- Fast classification
- Not sensitive to irrelevant features
- Can handle real-valued data

- **Disadvantage:**

- Assume independence of features